

Literature-implied partial answer: automorphic separable $C(K)$ spaces

Run `fa_banach_001`; source arXiv:1701.04263; supporting arXiv:0903.0658

Status

literature_implied_answer (partial subcase).

Anisca–Ferenczi–Moreno, arXiv:1701.04263, ask in Question 1.1 whether c_0 and ℓ_2 are the only separable automorphic Banach spaces. This note records a known-literature implication for the subcase of separable spaces of the form $C(K)$.

Claim

Let K be compact metrizable. If $C(K)$ is infinite-dimensional and automorphic, then $C(K)$ is isomorphic to c_0 .

Thus the answer to Question 1.1 is affirmative inside the class of separable $C(K)$ -spaces: no non- c_0 separable $C(K)$ -space is automorphic.

Identification

Every compact metrizable K is a continuous image of the Cantor set, and the Cantor set is Valdivia compact. Aviles–Moreno prove that if K is a continuous image of a Valdivia compact and $C(K)$ is extensible, then $C(K)$ is isomorphic to $c_0(\Gamma)$. Their introduction also recalls the Moreno–Plichko fact used in the area: automorphic spaces are extensible. Consequently an automorphic $C(K)$, with K compact metrizable, is isomorphic to $c_0(\Gamma)$. Since $C(K)$ is separable, Γ is countable in the infinite-dimensional case, and therefore $C(K) \simeq c_0$.

The supporting authors explicitly address the $C(K)$ case of the general automorphic-space problem, but arXiv:0903.0658 predates arXiv:1701.04263. The connection to Question 1.1 of arXiv:1701.04263 is therefore recorded here as an agent-identified literature implication, not as a new theorem and not as a later exact answer to that paper.

Evidence

The source question appears in arXiv:1701.04263, page 3, Question 1.1. The supporting theorem is arXiv:0903.0658, page 2, Theorem 1. The supporting paper’s abstract and introduction state the same $C(K)$ -automorphic conclusion on page 1.

The packet includes local copies of both PDFs:

`source_paper.pdf`, `supporting_paper_0903.0658.pdf`.

Scope

This is only a $C(K)$ -subcase. It does not settle the full question for arbitrary separable Banach spaces, does not address the UFO classification problem, and does not answer the separate question in arXiv:1701.04263 about reducing E_0 to positions inside $C(0, 1)$.

Search Notes

Local run searches for arXiv:1701.04263, automorphic spaces, UFO/uniformly finitely extensible spaces, $C(K)$, $C[0, 1]$, and position keywords found no prior packet for this subcase. Web searches on 2026-06-15 found the source paper and arXiv:0903.0658, but no later full resolution of the broad automorphic-space question.

References

- [1] R. Anisca, V. Ferenczi, and Y. Moreno, *On the classification of positions and of complex structures in Banach spaces*, arXiv:1701.04263.
- [2] A. Aviles and Y. Moreno, *Automorphisms in spaces of continuous functions on Valdivia compacta*, arXiv:0903.0658.